Software Quality Assurance

Important Short Questions (15) · 2-mark answers

Learn first (★★★)	Then read once (★★)
Q1 Defect · Q2 Handling a defect · Q6 Process vs Product · Q7 Internal vs External · Q8 Quality metric · Q9 Correctness vs Reliability	Q3, Q4, Q5, Q10, Q11, Q12, Q13, Q14, Q15

Topics already in the long questions (QA vs QC, Cost of Quality, Benchmarking, Garvin, Dimensions, McCall/ISO, Basic/Expected/Exciting) are not repeated here. If one of them comes as a short, write 3–4 points from its long-question table.

PPT 1: Quality

Q1. What is a defect? ★★★ (Slide 19)

A defect is the result of a **mistake** made during development. It introduces a **fault** in the software. Missing, unclear or wrong **requirements** are also defects.

Error → Defect → Fault → Failure

Example: a developer writes + instead of − in a bill calculation, so the customer gets a wrong total.

Q2. What is the best way to handle a defect? ★★★ (Slide 20, 22)

The best and cheapest way is to **prevent** the defect. The **later** a defect is found, the **more** it costs to fix: **1 unit** in requirements can become **100+ units** after delivery.

Example: fixing a wrong requirement on paper is cheap, but fixing it in released software is very costly.

Q3. What is quality? ★★ (Slide 3–4, 10)

There is **no single definition** of quality. Quality means **fitness for use**. According to ISO, quality is the degree to which a system **meets specified requirements** and customer needs.

Example: a calculator app is of good quality if it gives correct answers and is easy to use.

Q4. Differentiate between quality of design and quality of conformance. ★★ (Slide 5)

Quality of Design	Quality of Conformance
Products are made in different grades **on purpose**	How well the product **matches its specification**
Example: cheap and expensive phones	Example: phone works exactly as designed

Q5. What are the benefits of software quality? ★★ (Slide 17–18)

- **Fewer defects** and less rework
- **Time and cost saving**
- **Higher reliability** and lower maintenance cost
- **Higher user satisfaction**

Example: an app with fewer bugs gets better reviews and needs fewer updates.

PPT 2: Software Quality Attributes

Q6. Differentiate between process quality and product quality. ★★★ (Slide 4)

Process Quality	Product Quality
Measure of the **activity** that makes the product	Measure of the **product itself**
Example: productivity, timeliness	Example: software, specifications, plans

A **quality process leads to a quality product**.

Q7. Differentiate between internal and external quality attributes. ★★★ (Slide 7)

Internal Attribute	External Attribute
Found from the **description**; system need not run	Found when the system **runs**
Example: Efficiency	Example: Reliability, Robustness

Internal quality **leads to** external quality.

Q8. What is a quality metric? ★★★ (Slide 8–9)

A metric is a **measurement scale** for a quality attribute, along with a **method** to measure it.

- **Internal metric:** code complexity
- **External metric:** testing of functions, reliability, performance

Example: number of errors per 1000 lines of code (errors/KLOC).

Q9. Differentiate between correctness and reliability. ★★★ (Slide 24–25)

Correctness	Reliability
Works **according to the specification**	**Probability** of working properly over time
Absolute: cannot be "almost correct"	**Relative**: e.g. fails once in 3 weeks

Q10. What is a quality attribute? ★★ (Slide 5)

A quality attribute is a **property of a process or product** that can be **measured or observed**.
Examples: maintainability, portability, usability.

Q11. Define software quality. ★★ (Slide 2)

Software quality is conformance to: (1) stated **functional and performance requirements**, (2) documented **development standards**, and (3) **implicit characteristics** expected of all professional software.

Example: a banking app must do correct transactions, follow coding standards, and be secure.

Q12. What are the barriers to achieving quality? ★★ (Slide 10–11)

- Not understanding the **importance** of quality
- Poor **languages and methods** to specify quality
- **Difficulty** in designing for quality
- **Quality control as an afterthought**

Example: a team tests the software only at the end, so many defects are found late.

Q13. Differentiate between availability and recoverability. ★★ (Slide 32–33)

Availability	Recoverability
Proportion of time the system is **up and running**	System **starts again from where it stopped** after a crash
Example: website up 99% of the time	Example: MS Word recovers a document

Q14. What is maintainability? ★★ (Slide 39–40)

Maintainability is how easy it is to **change** a system after its release. Its sub-qualities are **Analysability, Changeability, Stability and Testability**.

Example: adding a new payment method to a shopping app easily, without breaking other features.

Q15. What are the process quality attributes? ★★ (Slide 47–49)

- **Productivity:** work done with given resources
- **Timeliness:** delivering **on time**
- **Visibility:** process and project status are **clear** to everyone

Example: a team delivers the app before the deadline (timeliness) and shares progress on a task board (visibility).

How to write a short answer

- Start with a clear **definition**.
- If the question says "differentiate", draw a **table**.
- Add **one example**.
- **Underline** the keywords.